ARTIFICIAL NEURAL NETWORKS AND DEEP LEARNING · FINAL PROJECT

Depression-Related Facial Expression Classification

A CNN trained from scratch vs. transfer learning with MobileNetV2 and ResNet50

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INTRODUCTION

Motivation and research question

Motivation

People with depression show fewer positive and more negative or flat facial expressions (Cohn et al., 2009; Girard et al., 2014).

Task

Classify one face image into the 7 basic emotions, and into 3 depression-related patterns: negative, neutral and positive affect.

Scope

The models recognise expressions linked to depression. They do not diagnose depression.

RESEARCH QUESTION

How does a CNN trained from scratch compare with transfer learning (MobileNetV2, ResNet50) for facial expression classification, in accuracy and efficiency?

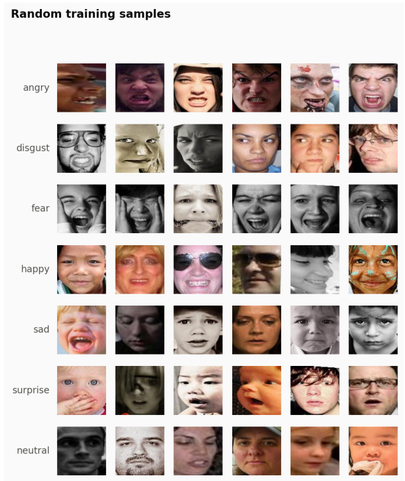
DATA

Two public datasets



FER-2013

- 35,887 images
- 48×48 grayscale
- 1 annotator per image: noisy labels
- Largest class: happy, 25.0%
- Smallest class: disgust, 1.5%



RAF-DB

- 15,339 images
- 100×100 colour, aligned
- ~40 annotators per image: reliable labels
- Largest class: happy, 38.8%
- Smallest class: fear, 2.3%

Same 7 emotions in both. Official test set kept, 10% of training held out for validation. Strong class imbalance, so we use class weights and report macro-F1.

DATA

Two label schemes

emotion7 keeps the 7 basic emotions. **affect3** groups them into depression-related expression patterns:

Negative

angry · disgust · fear · sad

46.5% of FER-2013 · 29.7% of RAF-DB

Neutral

neutral

17.3% of FER-2013 · 20.9% of RAF-DB

Positive

happy · surprise

36.2% of FER-2013 · 49.4% of RAF-DB

Depression is associated with more negative and less positive facial affect, so affect3 is our depression-related task.

METHOD

Three models, one pipeline

	Scratch CNN	MobileNetV2	ResNet50
Pretrained	No	ImageNet	ImageNet
Input	Native: 48×48×1 or 100×100×3	224×224×3	224×224×3
Architecture	4 conv blocks (Conv–BN–ReLU ×2, pool, dropout) + small head	Inverted residual blocks, depthwise convolutions	50-layer residual network
Parameters	4.8M	2.3M	23.6M
Training	From scratch, learning rate 10^{-3}	Two-phase fine-tuning	Two-phase fine-tuning

Shared by all models: the same splits, augmentation (flip, rotation, zoom, shift), class weights, Adam, early stopping and test set.

METHOD

Two-phase fine-tuning

PHASE 1

Train the new head

- ImageNet backbone frozen
- Only the new head learns (9k–14k parameters)
- Learning rate 10^{-3}, up to 10 epochs

PHASE 2

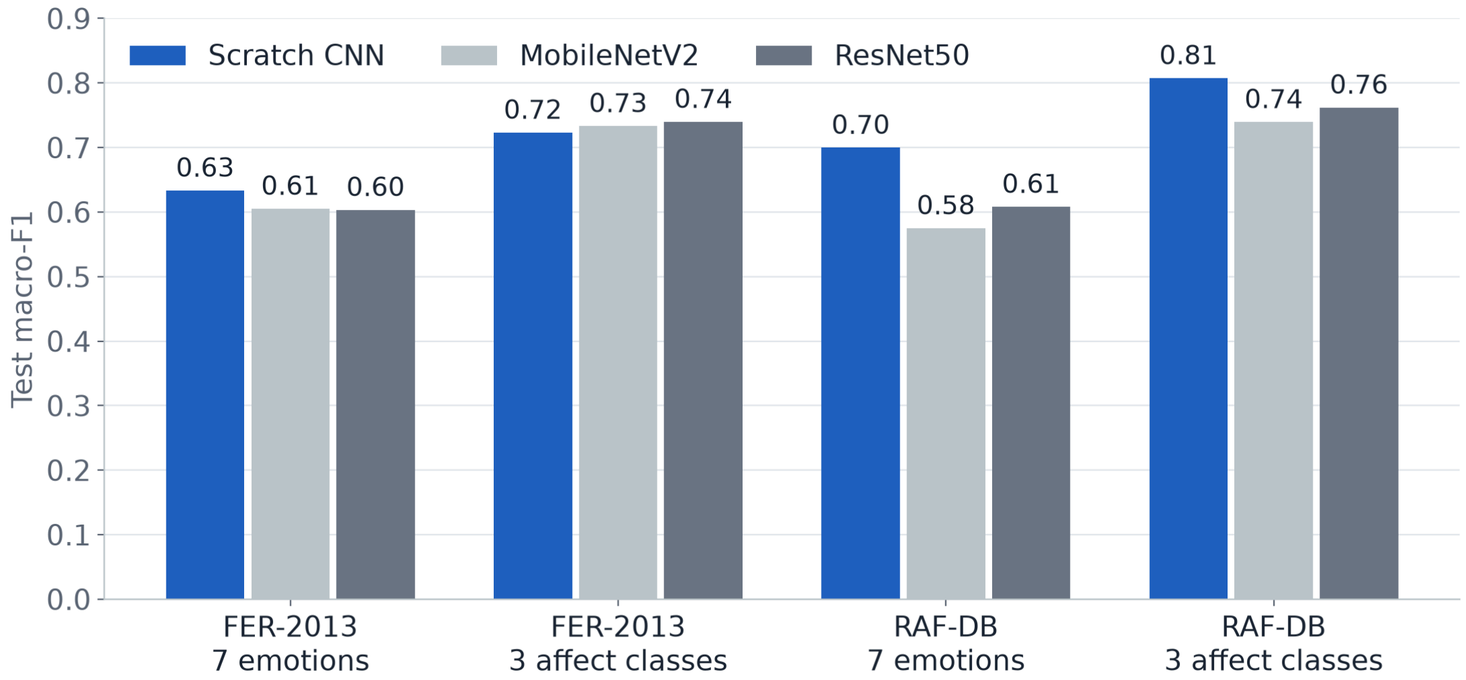
Fine-tune the top stage

- Unfreeze MobileNetV2 blocks 13–16 or ResNet50 conv5
- 1.7M or 15.0M trainable parameters
- Learning rate 10^{-5}, BatchNorm kept frozen
- Early stopping on validation loss

Model: 224×224×3 face → ImageNet preprocessing → pretrained backbone → global average pooling → dropout 0.3 → softmax.

RESULTS

Test results: macro-F1



3 of 4

settings where the scratch CNN has the best macro-F1

+4 to +13

accuracy points for the scratch CNN on RAF-DB

< 2

points: ResNet50's lead on FER-2013 affect3, the one setting it wins

RESULTS

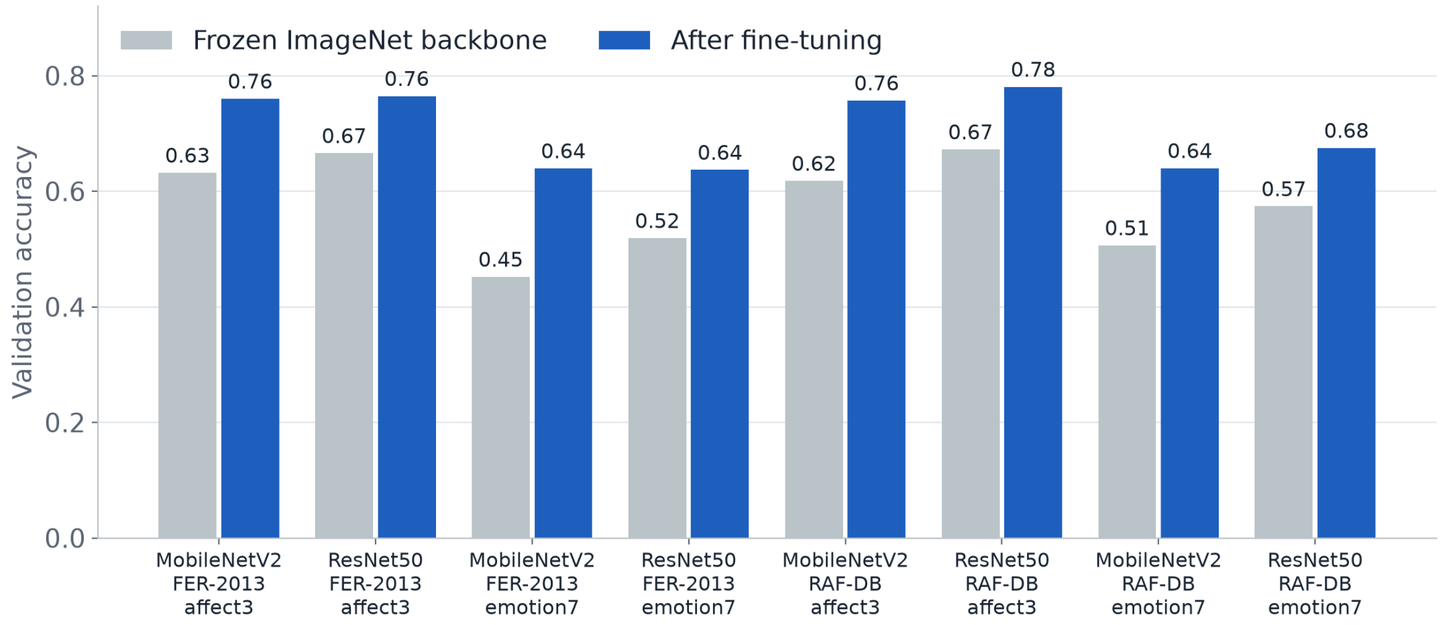
Accuracy and macro-F1 on the test sets

Setting	Scratch CNN	MobileNetV2	ResNet50	Best macro-F1
FER-2013 · 7 emotions	64.8% / 0.633	63.0% / 0.605	63.8% / 0.603	Scratch CNN
FER-2013 · 3 affect	74.5% / 0.723	76.0% / 0.733	76.3% / 0.740	ResNet50
RAF-DB · 7 emotions	78.7% / 0.700	65.4% / 0.575	69.7% / 0.608	Scratch CNN
RAF-DB · 3 affect	82.4% / 0.808	75.6% / 0.740	78.1% / 0.762	Scratch CNN

Each cell: test accuracy / macro-F1, one run per setting (seed 42). The RAF-DB 7-emotion scratch CNN used tuned settings (learning rate $3 \cdot 10^{-4}$, up to 100 epochs); all other runs used the same defaults.

RESULTS

Effect of fine-tuning



45–67%

validation accuracy with frozen ImageNet features

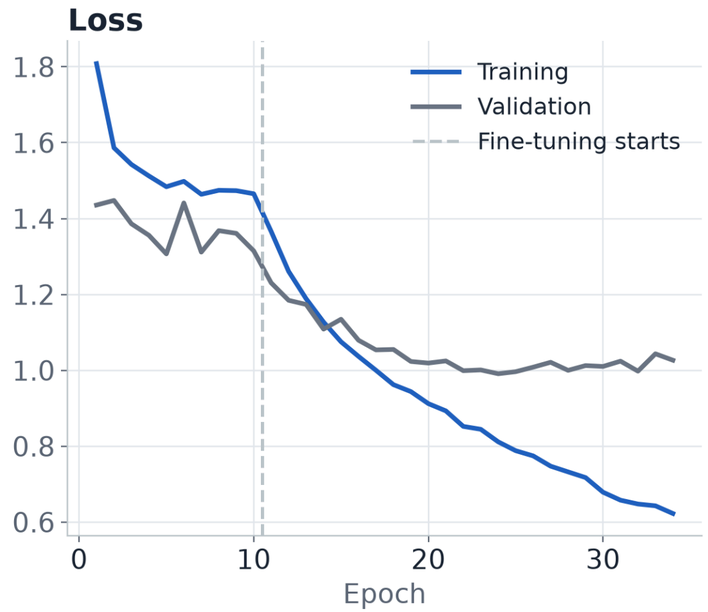
+10 to +19

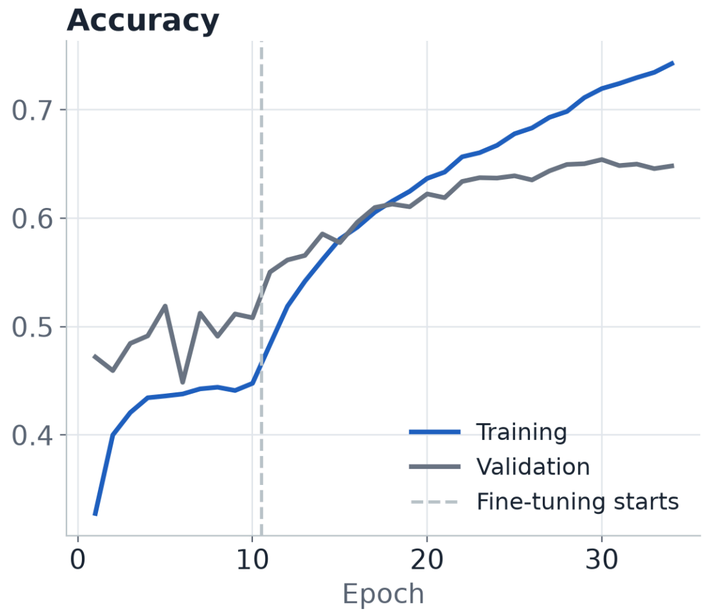
points gained by fine-tuning, in every setting

ImageNet features are not ready-made face features: they need substantial adaptation.

RESULTS

Training curves: ResNet50 on FER-2013





Frozen backbone

Slow progress, validation accuracy about 0.5

Fine-tuning

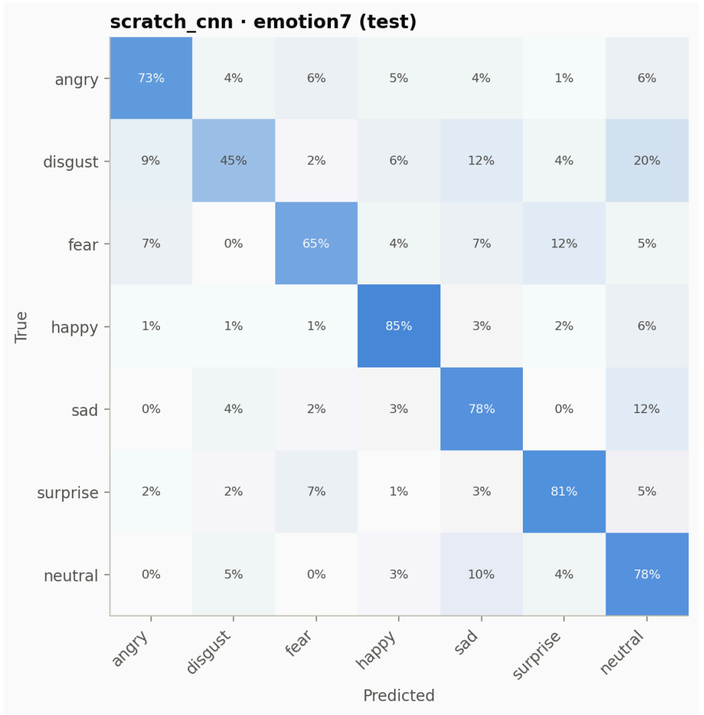
Clear jump once the top stage is unfrozen

Overfitting

Training accuracy 0.74 vs validation about 0.65; early stopping at epoch 34

RESULTS

Where the models make mistakes



RAF-DB · scratch CNN · confusion matrix



FER-2013 · scratch CNN · correct (top) and incorrect (bottom), true → predicted

Most common confusions

- disgust → neutral (20%), fear → surprise (12%), sad → neutral (12%)
- Depression-related classes: errors are mostly negative ↔ neutral

RESULTS

Efficiency

Scratch CNN

0.18 ms

per image (FER-2013)

4.8M

parameters

MobileNetV2

1.15 ms

per image (FER-2013)

2.3M

parameters

ResNet50

2.33 ms

per image (FER-2013)

23.6M

parameters

Inference on the Colab GPU (RAF-DB: 0.64 / 1.31 / 2.48 ms). The scratch CNN is 2–13× faster because it reads small native images; ResNet50 has 10× the parameters of MobileNetV2 for at most 3 macro-F1 points.

MAIN FINDING

On small face datasets, a CNN trained from scratch matched or beat ImageNet transfer learning, and it was faster.

Likely reasons: the gap between ImageNet objects and faces, fine-tuning only the top stage, ResNet50 overfitting, and 48×48 faces upscaled to 224×224.

DISCUSSION

Limitations and future work

Limitations

- One run per setting (seed 42), so no error bars
- Planned extra seeds and 10–50% data runs not done in the available GPU time
- One scratch CNN run had tuned settings
- Only the top backbone stage was fine-tuned
- No clinical labels: not a diagnostic tool

Future work

- Several seeds and less training data, to see when pretraining helps
- Fine-tune more layers or the whole backbone
- Smaller inputs (96–128 px) for the pretrained models
- Backbones pretrained on faces instead of ImageNet
- Video and clinically labelled data; fairness checks

DISCUSSION

Conclusion

3 of 4 settings won by the CNN trained from scratch, which was also the fastest model

+10 to +19 points from fine-tuning: transfer learning only works once the backbone adapts to faces

82.4% best accuracy on the depression-related negative / neutral / positive task (RAF-DB, scratch CNN)

Thank you. Questions?

github.com/SaraBedulla/FacemoodClassifier